



Lecture

10-12

Estimation and Types



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Estimates

- Estimates are used to **predict values** that are not yet known.
- Estimation is all about **predicting costs, revenues, time, and resources** for engineering projects before they actually happen.

Types of Estimates:

Depending on the context and the level of information available:

- ☐ Rough Order of Magnitude (ROM) Estimate
- ☐ Budget Estimate
- ☐ Definitive Estimate
- ☐ Parametric Estimate
- ☐ Analogous Estimate
- ☐ Bottom-Up Estimate



Rough Order of Magnitude (ROM) Estimate

- It is also known as a **ballpark estimate**.
- A very high-level estimate made early in the project **planning phase**. It provides a **rough idea** of the cost or effort involved, often with a wide range of **uncertainty**.
- It is a preliminary and approximate assessment of the
 - Cost
 - Time
 - Resources required for a project, task, or endeavour.
- A general sense of the magnitude of these factors without going into detailed analysis or calculations.
- ROM estimates are typically used in the **early stages of project planning** when there is **limited information** available and when more accurate estimates would be impractical or premature.
- A ROM estimate is usually accurate within: **-50% to +100%**



Few key points → ROM estimates:

➤ **Approximate Nature:**

- **Deliberately imprecise**
- Often given as a **range** (e.g., 1,00,000 to 2,00,000) **rather than a specific value.**
- Acknowledges the **uncertainty and variability inherent** in the early project stages.

➤ **Limited Detail:**

- Estimates are based on *high-level information* and *assumptions*.
- They don't take into account the full scope, specific requirements, or potential risks that could impact the project.

Quick Assessment: ROM estimates are used to quickly gauge the feasibility of a project or to provide a **ballpark figure for budgeting and decision-making without investing a lot of time or effort into detailed planning.**

Communication: ROM estimates **are useful for communicating with stakeholders, clients, or team members to give them an initial idea of what to expect in terms of cost, time, or effort.**



Here are a few key points about ROM estimates:

- **Basis for Further Planning:** As the project progresses and more information becomes available, ROM estimates can be refined and replaced with more accurate estimates based on detailed analysis and specifications.
- **Subject to Change:** Since ROM estimates are based on limited information and assumptions, they are subject to change as more data becomes available or as the project's scope and requirements evolve.
- It's important to note that while ROM estimates are valuable for initial planning and decision-making, they should not be used as the sole basis for making critical project decisions.
- As the project advances and more details are gathered, it's essential to replace ROM estimates with more accurate estimates to ensure successful project execution.



Examples

If someone asks:

“How long does it take to travel from Delhi to Mumbai?”

A ROM answer: “Around 1–2 hours by flight”, “Around 20–25 hours by train”

You didn’t check schedules — you just used experience. That’s ROM.

Build a basic college web application

High-level costs (rough):

Development (1 student, ~1 month) → ₹25,000

Cloud hosting (basic server) → ₹2,000

Domain & tools → ₹2,000

ROM Budget:

👉 ~₹30,000

Accuracy range:

₹15,000 – ₹60,000

Key point: **ROM estimates are used early for decision-making, not exact budgeting.**



Example 1: Construction Project

Example:-You are tasked with estimating the cost of constructing a new office building. Based on your experience and knowledge, you estimate that the cost per square foot for construction will be around 150 to 200 INR. The building is expected to have a total area of 10,000 square feet. Calculate the rough order of magnitude cost estimate.

Solution:

Average cost per square foot = $(150 + 200) / 2 = 175$

Total cost estimate = Average cost per square foot * Total area = $175 * 10,000 = 1,750,000$ INR

So, the ROM estimate for the construction of the office building is approximately 1,750,000 INR.



Example 2: Software Development Project

Example:- You're leading a software development project and need to estimate the duration of the project. Based on your knowledge, similar projects have taken between 6 to 9 months to complete. Estimate the rough order of magnitude duration for your project.

Solution:

Average duration = $(6 + 9) / 2 = 7.5$ months

So, the ROM estimate for the duration of the software development project is approximately 7.5 months.



Example 3: Product Manufacturing

Example:- You're working on estimating the production cost for a new electronic gadget. Based on your understanding of the components and manufacturing processes involved, you estimate that the cost per unit will be between 50 and 70 INR. You anticipate producing around 5,000 units. Calculate the rough order of magnitude cost estimate.

Solution:

Average cost per unit = $(50 + 70) / 2 = 60$ INR

Total cost estimate = Average cost per unit * Number of units = $60 * 5,000 = 3,00,000$

The ROM estimate for the production cost of the electronic gadgets is approximately 3,00,000 INR.

Remember, the key to ROM estimates is to provide **a broad, preliminary idea of the potential costs or timelines.** These estimates can be refined and made more accurate as more detailed information becomes available during the planning and development phases.



Budget Estimate

➤ Budget Estimate:

- More refined than a ROM estimate but is still based on limited information.
 - Used for **preliminary budgeting purposes**.
 - Preliminary calculation or **approximation of the costs** associated with a project, task, or plan.
 - An early assessment of the financial resources required to complete the project or achieve a certain goal.
 - **Budget estimates are typically created before detailed planning takes place and serve as a starting point for the decision-making and resource allocation.** They are useful in determining the **feasibility** of a project and setting initial funding limits.



Budget Estimate -- Construction Project:

Example:-Imagine a construction company planning to build a new office building. Before they start developing detailed architectural plans, they need a rough idea of the potential costs involved. They create a budget estimate that includes factors such as **land acquisition, construction materials, labor costs, permits, and other expenses**. This initial estimate helps the company decide whether the project is financially viable and secure funding from stakeholders.

A construction company is estimating the budget for a new building project. The estimated costs are as follows:

Materials: 2,00000 INR

Labor: 3,00000 INR

Equipment: 1,00000 INR

Overhead (10% of total direct costs): 60,000 INR

Total Estimated Budget = Materials + Labor + Equipment + Overhead

Total Estimated Budget = 2,00000 + 3,00000 + 1,00000 + 60,000

Total Estimated Budget = 6,60,000 INR



Product Development Budget Estimate

Example:- A tech company is planning to develop a new software product. The estimated costs are as follows:

Research and Development: 1,50,000

Testing and Quality Assurance: 80,000

Marketing and Promotion: 50,000

Miscellaneous Expenses: 20,000

Total Estimated Budget = R&D + Testing + Marketing + Miscellaneous

Total Estimated Budget = 1,50,000 + 80,000 + 50,000 + 20,000

Total Estimated Budget = 3,00,000 INR

These examples showcase different scenarios where budget estimates are calculated for various engineering and business projects. Keep in mind that these figures are fictional and simplified for illustration purposes.

Real-world budget estimates can be much more complex and involve additional factors like inflation, interest rates, and project duration.



Definitive Estimate

- The **most accurate estimate** and is typically produced later in the project lifecycle when detailed information is available. **Accuracy: typically $\pm 5\%$ to $\pm 10\%$**
- It involves a **comprehensive breakdown of costs and resources**.
- It is typically produced when the project scope is well-defined, detailed plans and specifications are available.
- It is prepared with a **high degree of accuracy** and details.
- There is a **high level of certainty** about the project's requirements and execution.



Definitive Estimate

Key characteristics:

- **Detailed Scope:** The project scope is clearly defined and documented, leaving little room for ambiguity or uncertainty.
- **Comprehensive Plans and Specifications:** Detailed engineering plans, drawings, and specifications are available, providing a clear understanding of the project's technical requirements.
- **Accurate Quantities:** Quantities of materials, labor, and other resources needed for the project are accurately determined based on the detailed plans and specifications.
- **Accurate Cost Breakdown:** The estimate breaks down costs into various components, such as materials, labor, equipment, overhead, and contingencies, with each component being well-defined and supported.
- **Reliable Data:** The estimate is based on reliable data, including current market prices for materials and labor, as well as historical cost data from similar projects.



Limited Assumptions: Assumptions made in the estimate are minimal and well-documented. The estimate is **largely driven by actual data** and concrete information.

Low Level of Risk: The level of uncertainty and risk associated with the estimate is minimal due to the detailed nature of the plans and specifications.

High Accuracy: Definitive estimates are expected to have a high degree of accuracy, typically within a narrow range (e.g., within 5% of the actual costs).

Approval and Funding: Definitive estimates are often used to **secure project funding** and obtain necessary approvals from stakeholders due to their high reliability.

Final Stage: Definitive estimates are typically prepared during the final stages of project planning, after the project scope and requirements have been fully defined.



- It's important to note that while a definitive estimate provides a high level of accuracy, it also requires a significant amount of time, effort, and resources to prepare.
- Additionally, it may not be suitable for projects with uncertain or evolving scopes, as changes to the scope could significantly impact the accuracy of the estimate.



Example 1: Enterprise Resource Planning (ERP) System Implementation

Example 1: A software company is tasked with implementing a comprehensive ERP system for a large manufacturing client. The project involves integrating various modules like *finance, inventory, procurement, production, and human resources*. **The company follows a detailed estimation process:**

Scope Definition: The project team collaborates with the client's stakeholders to define the scope, requirements, and functionalities of the ERP system. A detailed list of features and modules is compiled.

Work Breakdown Structure (WBS): The project team creates a comprehensive WBS, breaking down the project into *smaller tasks and sub-tasks*. Each module and feature is allocated a specific task or set of tasks.

Resource Estimation: The company identifies the human resources, equipment, software licenses, and infrastructure required for the project. *This includes developers, QA engineers, project managers, hardware, and software.*



Cost Estimation: Based on the resource requirements, the company estimates costs for *labor, equipment, licenses*, and any other *project-related expenses*. **Vendor quotes are obtained for software licenses and hardware procurement.**

Risk Analysis: Potential risks are identified, analyzed, and quantified. Contingency plans are devised to mitigate these risks, and additional costs are factored into the estimate.

Scheduling: A detailed project schedule is created, taking into account task dependencies, resource availability, and milestones. The Gantt chart illustrates the project timeline.

Quality Assurance: The QA process is outlined, detailing the types of testing (functional, integration, performance) and the resources required for thorough testing.

Client Review: The comprehensive estimate is presented to the client for review and approval. Any discrepancies or concerns are addressed in collaboration with the client.

Final Estimate: **After incorporating client feedback, the company provides a definitive estimate that includes accurate costs, a detailed project schedule, and a breakdown of resource allocations.**



Example 2: Mobile App Development for E-Commerce Startup

Example 2: A software company is hired to develop a mobile app for a startup e-commerce business. The app should support **both iOS and Android platforms** and include features like product catalog, user authentication, shopping cart, and payment integration.

Requirement Analysis: The project team works closely with the startup's stakeholders to gather and analyze requirements. Detailed user stories and use cases are documented.

Platform Selection: The company evaluates technology stacks for both iOS and Android development, considering factors like language (Swift/Kotlin), framework, and third-party libraries.

Feature Breakdown: Each feature of the app is broken down into tasks, such as UI design, backend development, API integration, and testing.

Resource Allocation: The company allocates designers, developers, QA engineers, and project managers to the project. Hardware and software requirements are identified.

Cost Calculation: Using hourly rates for each resource and considering the estimated time for each task, the company calculates the overall cost of development.

Timeline Creation: The project timeline is created, highlighting key milestones such as UI design completion, backend development completion, and testing phases.



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Testing Strategy: The testing plan is outlined, detailing how different aspects of the app will be tested, including functionality, performance, and security.

Client Collaboration: The company collaborates closely with the startup to ensure alignment with the evolving business needs and to make any necessary adjustments to the project plan.

Risk Management: Potential risks like technical challenges or delays are identified, and contingency plans are developed. These plans may involve additional resources or schedule adjustments.

Client Approval: The definitive estimate is presented to the startup, allowing them to review costs, timelines, and project details. Feedback is incorporated into the final estimate.

Final Proposal: The software company provides the startup with a comprehensive definitive estimate that includes a detailed breakdown of costs, timelines, resource allocation, testing plans, and risk management strategies.



Types of Estimate -- Parametric Estimate

- **Parametric estimating uses a statistical relationship between variables to calculate the estimate.** It's particularly useful when estimating based on **similar past projects**.
- Cost or time estimation technique used in project management, engineering, and various other fields.
- Based on a set of parameters or variables that are known to influence the cost or time of a project, task, or activity. **These parameters could include factors such as size, complexity, resources, and historical data.**
- Established a mathematical relationship between the parameters and the project's cost or duration. This relationship is often derived from historical data or industry benchmarks.
- **Once the relationship is established, it can be used to calculate estimates for new projects or tasks.**



Example:

➤ **Let's say you're estimating the cost of building a house. You might find that the cost per square foot for a house in your area is 200, and you know that the house you're planning to build is 2,000 square feet. Using a parametric estimate, you can calculate the cost as:**

Cost = Cost per square foot × Square footage

Cost = 200 × 2,000 = 4,00000

- The cost per square foot is the parameter, and the square footage is the value of that parameter for the specific project.
- It can be useful when you have limited data or time to perform a detailed analysis. However, they rely on the assumption that the relationship between the parameters and the cost or time is consistent and applicable to the new project.
- It's important to note that while parametric estimates can provide quick and rough estimates, they might not be as accurate as more detailed estimation methods, especially for projects with unique characteristics or those that significantly deviate from the historical data used to establish the parameters.



Example

A manufacturing company wants to estimate the machining cost of a batch of steel shafts. From historical data, the average machining cost is ₹180 per kg of finished shaft. The company plans to manufacture 250 shafts, and each shaft weighs 2.5 kg. Using parametric estimation, calculate the estimated total machining cost.



Parameter = Weight of finished product

Cost per kg = ₹180/kg

Total weight:

$$250 \times 2.5 = 625 \text{ kg}$$

Estimated machining cost:

$$625 \times ₹180 = ₹1,12,500$$



Example

An electronics company wants to estimate the assembly cost of PCBs for a new product. Based on historical manufacturing data, the average PCB assembly cost is ₹35 per component mounted. The new product requires: 1,200 PCBs Each PCB has 80 components. Using parametric estimation, calculate the estimated total PCB assembly cost.



Parameter = Number of components

Total components:

$$1,200 \times 80 = 96,000 \text{ components}$$

Estimated assembly cost:

$$96,000 \times ₹35 = ₹33,60,000$$



Types of Estimate - Analogous Estimate

- Analogous Estimate (also called **top-down estimating**) is a technique where you estimate a project by comparing it to a similar past project and adjusting as needed.
 - **Analogous estimating uses the cost, time, or effort of a previous similar project to estimate the current one.**
 - It is a **project management technique** used to estimate the duration, cost, or other relevant metrics of a project by drawing comparisons with similar projects.
 - It can be a useful technique **when there's limited information available for the new project**, especially during the early stages of project planning.
 - It provides only a rough estimate and should be used in conjunction with other estimation techniques for more accurate results.
- ✓**Note:** For any estimation approach, the **accuracy** of the estimate depends on the **quality of the data**, the expertise of the individuals involved, and the degree of similarity between the reference project and the project being estimated.



Process Flow - Analogous Estimate

- **Identify a Reference Project:** **Identification** of similar project w.r.t. *nature, scope, and complexity*. Identified reference project should have historical data available viz. project duration, cost, resources etc.
- **Gather Data:** **Collection** all available data related to the reference project, including *project plans, budgets, schedules*, and any other documentation. The more detailed and accurate the data, the better your estimate is likely to be.
- **Analyze Similarities and Differences:**
Comparison with the reference project for *similarities* in terms of *size, scope, objectives, constraints* etc. Identify any *differences* that might affect the estimate e.g., changes in technology, market conditions, or regulatory requirements.
- **Adjustment Factors:** **Apply adjustment factors** to account for the differences between the reference project and the project being estimated. These factors could be based on *expert judgment* or *quantitative analysis*. e.g., if the new project has stricter regulatory requirements, you might add a percentage to the estimate to reflect the additional effort required for compliance.
- **Calculate Estimate:** **Calculation of Adjusted Estimated** duration, cost, or other metrics for the new project.



Example - Analogous Estimate

➤ **Suppose you are tasked with estimating the effort (in person-hours) required for a new software development project that involves building a content management system (CMS) similar to one you've previously worked on. The previous CMS project had the following characteristics:**

Project Size: 10,000 lines of code

Effort Required: 1,000 person-hours

Development Time: 3 months

Now, you have a new project to estimate, which involves building a similar CMS with the following characteristics:

Project Size: 15,000 lines of code (larger than the previous project)

Unknown Effort

Desired Development Time: 4 months



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➤ Using the analogous estimating technique, you can *estimate the effort* required for the new project based on the historical data from the previous project. The basic formula for analogous estimating is:

$$\text{Estimated Effort} = (\text{Previous Effort}) \times (\text{New Size}) / (\text{Previous Size})$$

$$\text{Estimated Effort} = (1,000 \text{ person-hours}) \times (15,000 \text{ lines}) / (10,000 \text{ lines})$$

$$\text{Estimated Effort} = 1,500 \text{ person-hours}$$

So, based on the historical data from the previous project, you estimate that the new project would require **approximately 1,500 person-hours** of effort to complete.

➤ **Note:** The analogous estimate **provides a rough estimate** based on similarities between projects. The accuracy of the estimate can vary depending on the degree of similarity between the projects, changes in technology, team experience, and other factors.



Types of Estimate - Bottom-Up Estimate

- Its commonly used in *software engineering and project management*.
- Bottom-up estimating is a technique where each small work activity is estimated individually, and then all estimates are added together to get the total project cost, time, or effort.
- It involves breaking down a project into smaller, more manageable components or tasks and estimating the effort and resources required for each individual task. These individual estimates are then aggregated to provide an overall estimate for the entire project.
- It offers increased accuracy.
- More time-consuming compared to other estimation techniques.
- It's important to strike a balance between the level of detail and the time available for estimation. Additionally, regular review and adjustment of estimates as the project progresses are essential for maintaining accuracy.



Bottom-Up Estimate process workflow

Task Identification: Identify and list all the tasks or components required to complete the project. These tasks should be as granular as possible, breaking down the project into smaller units of work.

Task Estimation: Estimate the effort, time, and resources required for each individual task. This estimation can be done using various techniques such as expert judgment, historical data analysis, or analogy with similar past projects.

Aggregation: Sum up the estimates to calculate the total effort, time, and resources required for the entire project. This gives you a more accurate estimate because it takes into account the specifics of each task.

Contingency: It's advisable to include a contingency buffer in the estimate to account for uncertainties, risks, and unexpected events that might arise during the project. This helps to ensure that the estimate remains realistic even if things don't go exactly as planned.

Validation: Review and validate the estimates with relevant stakeholders, including team members, project managers, and clients. This step helps to identify any discrepancies or potential issues early on.

Documentation: Document the estimation process, including the breakdown of tasks, individual estimates, assumptions, and the contingency buffer. This documentation serves as a reference point throughout the project and helps in tracking progress.



Example - Bottom-Up Estimate

Consider a hypothetical software engineering project: building a content management system (CMS) for a small business. The project will be break down using a bottom-up estimation approach.

Step 1: Identify Tasks

First, we need to identify the tasks required to complete the project. These tasks might include:

- Requirements gathering and analysis
- Database design
- User interface design
- Backend development
- Frontend development
- Testing and quality assurance
- Deployment and launch



Step 2: Estimate Task Efforts

Next, we estimate the effort required for each task. Let's consider the "Backend development" task as an example. The backend development task can be further broken down into sub-tasks:

- Setup project environment
- Design and implement database schema
- Develop API endpoints
- Implement user authentication
- Implement data storage and retrieval logic
- Error handling and validation
- Documentation
- For simplicity, let's assume we are estimating in hours and each sub-task takes a certain number of hours to complete.



Step 3: Summing Up Efforts

- Now we sum up the estimated efforts for all sub-tasks to get the total effort for the "Backend development" task. Let's say the total effort for the backend development task is 300 hours.
- Similarly, we estimate efforts for all the other tasks, such as frontend development, testing, and deployment.



Step 4: Summing Up All Tasks

Finally, we sum up the efforts of all the tasks to get the overall effort for the entire project. Let's assume the total efforts for all tasks are as follows:

Requirements: 40 hours

Database design: 20 hours

User interface design: 30 hours

Backend development: 300 hours

Frontend development: 250 hours

Testing and QA: 80 hours

Deployment and launch: 20 hours

The total effort for the project would be 740 hours.



Step 5: Contingency and Review

- It's important to include a contingency factor to account for unforeseen delays, changes, or additional tasks that may arise during the project. For example, **adding a 20% contingency buffer would increase the total estimated effort to 888 hours ($740 * 1.20$).**
- Remember that these estimates are based on assumptions and experience. Actual project timelines may vary due to factors such as team efficiency, external dependencies, and scope changes.
- Bottom-up estimation allows you to create a detailed and granular estimation by breaking down the project into smaller tasks. This approach can help in better resource allocation, project planning, and identifying potential bottlenecks early in the project lifecycle.